

METHODOLOGY

4 | Literature Search

The use of peer reviewed literature have had the highest priority. As mentioned in chapter 3 the database Scopus has been used to great extend. By searching for literature systematically according to the framework by Aalborg University Library it has been possible to gather sources that provide the study with information about both gentrification and machine learning. By using the methods 'boolean operators', which enables the search for multiple key terms and exclusion of key terms as well (Aalborg Universitetsbibliotek, 2025). The use of 'chain searching' has also been essential to find the main source of valuable information, which consists of research in the bibliography of relevant articles. By repeating this process it is possible to find the core references for a given topic. In the case of gentrification it is paramount, since the same theorists are being cited through time (AU Studypedia, 2025). Lastly the project has been assisted by the supervision team consisting of Jamal Jokar and Janni Sørensen, who have supplied the study with critical information and literature.

5 | Use of AI and Open Science

Use of AI

Artificial Intelligence (AI) has played a noteworthy part in the development of this project. The use of AI has been in line with the code of conduct regarding AI in study aspects and has exclusively been used in assisting with the development of the coding for the ground truth data and the ML model. The AI agents used for this matter is the Google Gemini Assistant and the GitHub Copilot from Microsoft, which is an application provided and approved by the AAU Library (Aalborg Universitetsbibliotek, 2025). GitHub Copilot has been used to convey the pseudo-code into usable prompts in the ML model. Google Gemini has been used for sparring and guidance when interpreting results and equations in the ML model. The agents have been used in line with the concept of having a 'human in the loop'. This means the agents have only been used critically and with both human verification and interference which ensures that the ML model is trustworthy and understandable (Wu et al., 2022). With the aim to be as transparent as possible a .md file have been created detailing the context that the AI has been given to understand and help in the development in the project. This file can be viewed in the GitHub repository with the file name "copilot_instructions.md".

Open Science

The concept of Open Science is the choice of making academic research and results accessible to the public. In this project transparency is aimed to contribute to the widespread movement countering the 'replication crisis'. Thus it is ensured that the research and coding can be reviewed critically and iterated in the future (Aarhus Universitet, n.d.).

Therefore, all the coding for the ground truth data and the ML model is accessible as external documentation in the GitHub repository. The valuation and classification of data mentioned in section 8.3.1 is described further in appendix A. The aforementioned .md file used to develop AI agents is also provided in the context of Open Science.

6 | Data and Materials

As stated in section 3.3 there are several drivers and measurements that are relevant in regards to gentrification. The data used is provided by The Danish Neighborhood Atlas (DNA) and encompasses a variety of socioeconomic parameters, which have been created for both municipal- and a neighborhood level. Because it is based on data from Statistics Denmark it has the robustness of being scalable and comparable countrywide and across municipalities. The provided data consists of 26 layers and covers elements of education, income, age, employment, housing and demographic changes. The relevance of the different layers are studied by accumulating the arguments stated in the main sources presented in the literature review. If a reference mentions the parameter as a factor that indicates gentrification, the parameter is marked with 'x' in the table below.

Table 6.1: Parameters used to measure gentrification, as mentioned in various studies

	Income	Education	Age	Housing prices	Public housing	Work	Family type	Crime	Residential mobility
Bredahl et al., 2021	x	x			x	x	x		
Chappel and Zuk, 2016	x		x	x		x		x	x
Cohen and Pettit, 2019	x	x	x	x		x	x	x	
Drew, 2018	x	x	x	x					
García, 2025	x	x	x	x	x		x	x	x
Knorr, 2019	x	x	x	x		x	x		
Larsen and Hansen, 2008	x	x	x		x	x		x	
Reades et al., 2019	x	x		x	x			x	
Vidal, 2019	x	x	x		x				
Yoo, 2023	x	x	x	x	x	x	x		x
Zuk et al., 2015	x	x	x	x	x	x	x	x	x
Sum	11	10	9	8	7	7	6	6	4

This study illustrates that all parameters are relevant to use in the ML model. The residential mobility has the least mentions but it is decided with four mentions that the parameter is still relevant. The following table 6.2 presents the parameters and their matching layers from DNA that will be used in the ML model. The data covers the years from 1990 to 2020 and consists of 20 relevant layers.

Table 6.2: Parameters with underlying layers that is used in the ML model

Parameter	Layer
Income	Income
Education	Primary and lower secondary Upper secondary school and vocational college Higher education
Age	18-25 years 26-40 years 41-55 years 56-69 years
Public housing	Public housing
Employment	Employed Unemployed Outside the labor market
Housing prices	Housing prices Price per square meter
Family type	Couples with children Couples without children Singles with and without children
Residential mobility	Inflow of residents Outflow of residents
Serious Crime	Serious crime

A few layers from the original DNA data collection are excluded i.e. the data parameter called 'Prosperity', which is left out of the model. 'Prosperity' is a calculated index that consists of the following data: Disposable income, serious crime, residents in public housing, highest education level and square meter price (Lange et al., 2023). This index is left out of the model to avoid redundancy and confusion. The data parameters 'Net migration rate' and 'Population count' are also left out, as the dataset has other proxy variables consisting of 'Inflow of residents' and 'Outflow of residents'.

Proximity data is also left out of the model. DNA has three parameters which covers the data types 'Distance to business academy', 'Distance to university' and 'Distance to university college'. This data is left out of the ML model, as the distance is measured from each cluster's centroid. This will give a distorted result, as each centroid is depended on the cluster's size as is thus unfit to use in this analysis (Lange et al., 2023).

The proximity data is also provided without the time aspect and thus do not illustrate the development through time. Furthermore, the proximity to renewed recreational areas and public transportation is relevant to measure gentrification, as mentioned in section 3.3. DNA does not provide this data and for this analysis it is chosen to leave out this data, as the static location of current areas does not shed light on how gentrification can happen over time.

The data from DNA has missing values or NoData in some cases. This is only a challenge when processing the 'Housing prices' and 'Price per m²' data. This is due to the lack of housing sales in the specific year and neighborhood that makes calculating the average price difficult. This is typically neighborhoods with a big amount of public houses, dormitories or nursing homes. The data is especially insufficient in some areas of Copenhagen in the period 1990-1992.

Each layer consist of clusters, also called neighborhoods, that are given a value corresponding to the layer theme for each year in the 30 year period. The clusters are generated by DNA from the population in the given area and the different road networks surrounding the cluster. Furthermore the clusters are bounded so that they do not exceed the municipal borders. The aim has been to make the clusters as small as possible while still complying with GDPR-regulations. Therefore, the clusters consist of a minimum of 100 residents per cluster and they are a minimum of 10,000m² (1 hektar) in size. The cluster division is consistent through the 30 year period. Clusters that have been sparsely populated at some point in the period needs to cover a bigger area to ensure the 100 resident-rule. Neighborhoods like this are visible in areas like Ørestaden or Carlsbergbyen where each cluster covers well over 100 residents today because of an drastically increase in population. Because of being relatively new neighborhoods these clusters have not been exposed to gentrification, also areas bigger than 500,000m² (50 hektars) are thus sorted out. Neighborhoods which have not seen a dramatic rise in housing and inhabitants are generally more relevant, when considering whether or not they will be gentrified.

The prediction model will be able to classify individual neighborhoods on a cluster-level of 100m x 100m. Therefore it is relevant to consider if a small scale neighborhood from DNA can be considered as gentrified. In this report the minimum level of gentrification to a single neighborhood will be in line with Suttles (1972) scale nr. 1, Block face, which is a close proximity, where children are permitted to play without supervision, one to two streets, as mentioned in section 3.1.1.

7 | Process for Developing the ML Model

The procedure for developing the ML model will be reviewed in general before being explained in depth in the following sections. A simplified flowchart covering the process is presented in figure 7.1.

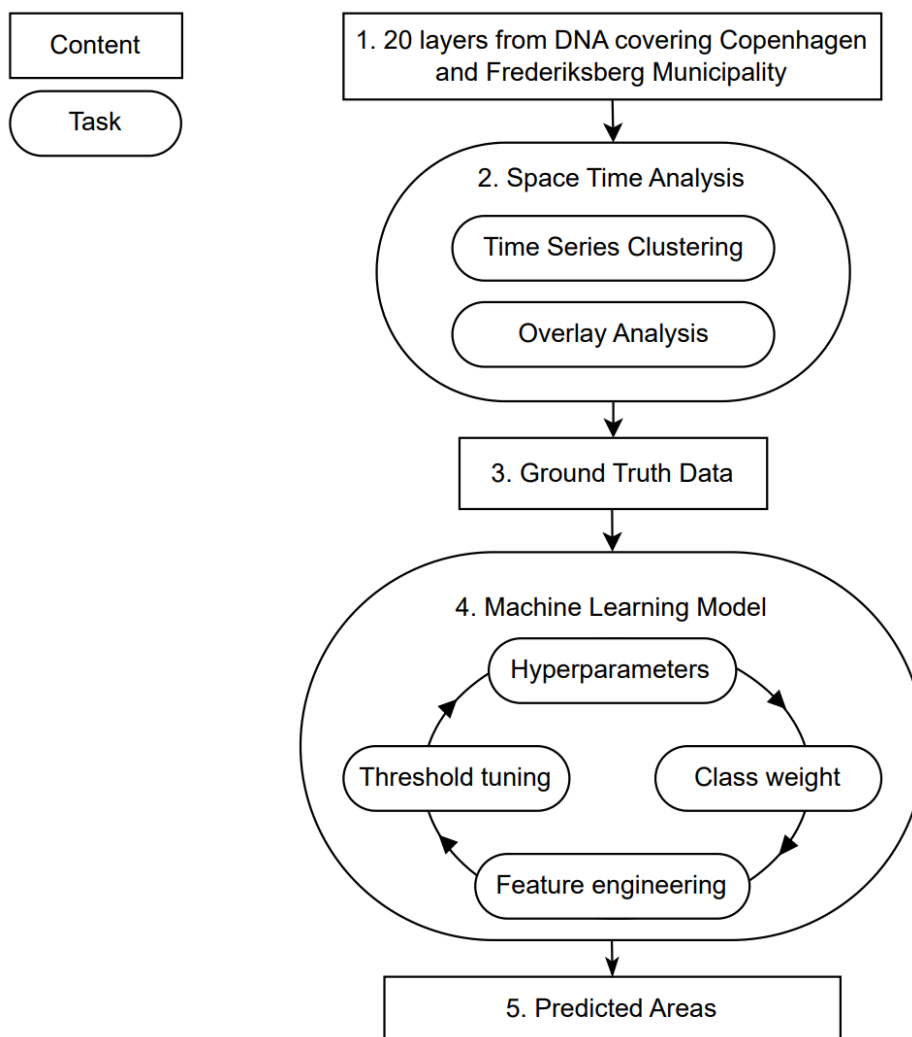


Figure 7.1: Workflow of the data processing

The 20 layers (step 1. in figure 7.1) corresponds to the layers mentioned in table 6.2. They are used in the space time analysis (2.). First it is divided into 'space time clusters' that are classified according to how clear signs of gentrification they show. A pairwise comparison

is conducted with six participants which results in a ranking of the parameters. This weighing is used in an overlay analysis together with the classified clusters to determine the ground truth data (GTD) (3.). The GTD features which areas that are gentrified, show no trends and not gentrified. Step 1.-3. is further uncovered in chapter 8.

The GTD is then used to train the ML model (4.). Different factors like the hyperparameters and thresholds are adjusted to further optimize the model. This process of step 3-5 is elaborated in chapter 9. When the accuracy is satisfactory the final result of predicted areas can be found and visualized by maps, which is presented in part III.

8 | Ground Truth Data

For a prediction model to be successful, it needs to have the proper training. This training-data needs to be as close to 100% certain as possible. However for the topic of gentrification, it is not as straight forward, since there is no clear definition of gentrification in the literature. Previous studies like Drew (2018) have compiled different approaches on to how to quantify gentrification, but most of these studies are hindered by limited data. The ML model developed in this project uses only data from The Danish Neighborhood Atlas (DNA). Since there are relevant 20 parameters to choose from, the task then becomes how to weigh each layer.

8.1 Space Time Analysis

This section will explain how the raw data from DNA is processed into Time Series Clusters, so that it can be reclassified to show which clusters show signs of gentrification.

Before making a ML model it is necessary to get an understanding of the chosen data from DNA and its patterns. With a Space Time Analysis (STA) the changes in data through time tied to a specific location is analyzed (ESRI, n.d.-b). The tool used to analyze these trends are Time Series Clustering in ArcGIS Pro. This tool gathers data defined by how the data for each location have been evolving over time. This shows which areas have undergone a similar change. Some neighborhoods might have been through a steady rise following the municipal or national mean, while others have seen a drastic surge at a specific period in time. The main aim of the analysis is to find areas that have been gentrified and can be used as ground truth data (GTD) in the machine learning model.

8.2 Scope of Analysis

In the limited literature on gentrification in Denmark, especially Inner Vesterbro stands out as gentrified but also Nørrebro, however to a slightly lesser extent. Therefore a scope is needed to capture the development of these neighborhoods. The area for the Space Time Analysis (STA) was the entire city of Copenhagen (Frederiksberg and Copenhagen municipal borders included). The map on figure 8.1 shows the borders for both the